AHB VERSION Sequence — Verification & Bug Report

Smart Home Alarm Controller | UVM Verification

1. Objective

The ahb_version_sequence performs an AHB read of the VERSION register. The ahb_version_test creates the UVM environment, starts the sequence on the AHB sequencer, and waits briefly after the read so the monitor and checker can process the transaction.

2. Sequence and Test Operation

1. **Create read sequence:** Create an ahb_read_sequence object named read_version.
2. **Set target address:** Set read_seq.read_addr = ADDR_VERSION (0x03C).
3. **Execute transaction:** Start the read sequence on m_sequencer.
4. **Observe transaction:** The AHB monitor captures the address and samples the returned HRDATA.
5. **Check readback:** The AHB checker compares the observed VERSION data against its expected value.
6. **Allow processing:** The test waits #20ns before dropping its UVM objection.

3. Expected and Observed Values

Item	Expected / requirement	Observed in simulation
Register address	0x0000003C	0x0000003C
Operation	Read (HWRITE = 0)	Read (HWRITE = 0)
Transfer type	Valid NONSEQ (HTRANS = 2'b10)	HTRANS = 2'b10
HREADY / HREADYOUT	Transfer accepted/completed	Both = 1
HRESP	OKAY = 0	0 (OKAY)
VERSION read data	0x00010008 (checker expectation)	0x00000000
UVM errors	0 expected for a passing test	1 UVM_ERROR

4. Evidence of the Failure

- At 18 ns, the AHB monitor reports: "Address captured: ADDR=0000003c WRITE=0 HTRANS=10".
- At 24 ns, the AHB monitor reports: "AHB READ ADDR=0000003c ... RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0".
- At 24 ns, the checker reports: "VERSION mismatch: EXPECTED=00010008 ACTUAL=00000000".
- The read sequence later reports: "READ ADDR=0000003c RDATA=00000000 RESP=0".
- The UVM summary reports 1 error, 0 warnings, and 0 fatal errors.

5. Waveform Evidence

The supplied waveform shows HADDR reaching 0x3C for the read, HWRITE low, HTRANS showing a valid NONSEQ transfer, HREADY and HREADYOUT high, HRESP low, and HRDATA remaining zero. The displayed internal acc_addr and acc_rdata traces also appear to remain zero in the screenshot. Because the screenshot does not clearly show the precise active access window for these internal signals, this is a clue rather than definitive proof that the address path is disconnected.

6. Bug Assessment

The VERSION register readback test fails: the checker expects 0x00010008 but the DUT returns 0x00000000. This is a confirmed readback mismatch. The project notes identify an injected VERSION encoding bug involving the order of VERSION_MINOR and VERSION_MAJOR in the concatenation. However, the observed zero value is not by itself

sufficient to prove that the encoding-order bug is the sole cause. The AHB-to-register access path should also be checked because acc_addr and acc_rdata appear to be zero in the waveform.

VERSION encoding / read-mux expression	Verify field ordering and width against the checker expectation.

7. Conclusion

ahb_version_sequence correctly issues a read to address 0x3C, and the monitor captures that transaction. The test exposes an incorrect readback value of zero and reports one UVM error. Therefore, the VERSION readback check fails. Further internal-signal inspection is needed to distinguish the known VERSION encoding defect from a problem in the AHB-to-register read-data path.

Appendix A — Complete Simulation Log

Complete log supplied with this test. The VERSION mismatch line is highlighted in red.

```
UVM_INFO @ 0: reporter [RNTST] Running test ahb_version_test...
UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0
alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in =
00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0
alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 18: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured:
ADDR=0000003c WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in =
00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB  READ ADDR=0000003c
WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing
transaction: ADDR=0000003c WRITE=0 RDATA=00000000
UVM_ERROR ahb_checker.sv(406) @ 24: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] VERSION mismatch: EXPECTED=00010008
ACTUAL=00000000
UVM_INFO ahb_checker.sv(467) @ 24: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB  READ: ADDR=0000003c HTRANS=10
HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 24: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=3c RDATA=0
UVM_INFO scoreboard.sv(27) @ 24: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB  READ: ADDR=0000003c DATA=00000000
RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 24: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=0000003c | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0
alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0
alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in =
00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 36: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_version [AHB_READ] READ
ADDR=0000003c RDATA=00000000 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0
alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in =
00000000
UVM_INFO alarm_reference_model.sv(231) @ 49: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync
Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0
alarm_active=0
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 56: reporter [TEST_DONE] 'run'
phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 56: reporter
[UVM/REPORT/SERVER]
--- UVM Report Summary ---
** Report counts by severity
UVM_INFO :   25
UVM_WARNING :    0
UVM_ERROR :    1
UVM_FATAL :    0
** Report counts by id
[AHB_ADDRESS]     1
[AHB_CHECK]     2
[AHB_MON]     1
[AHB_PUBLISH_DEBUG] 1
[AHB_READ] 1
[ALARM_DEBUG] 6
[REF_MODEL] 5
[RNTST] 1
[SCOREBOARD] 1
[SENSOR_AHB_DEBUG] 1
[SENSOR_MON] 4
[TEST_DONE] 1
[UVM/RELNOTES] 1
```

Appendix B — Waveform Evidence

Waveform captured for ahb_version_test. The image shows the AHB read at address 0x3C, HWRITE low, a valid NONSEQ transfer, HREADY and HREADYOUT high, HRESP low, and HRDATA at zero. The displayed acc_addr and acc_rdata traces also appear to remain at zero in this capture.

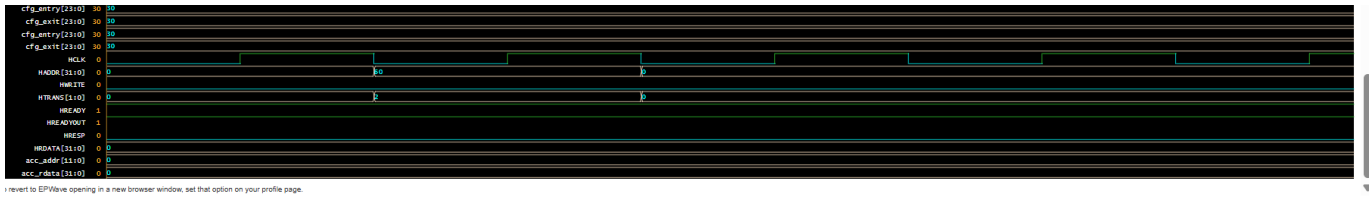


Figure B1. AHB VERSION register read waveform supplied for the failing test.

AHB Delay Configuration Sequence Verification Report

Smart Home Alarm Controller — UVM Verification

1. Objective

The objective of ahb_delay_config_sequence is to verify write and readback behavior for the ENTRY_DELAY and EXIT_DELAY registers through the AHB interface. The sequence also checks that updating one delay register does not incorrectly update the other.

2. Register Information

Register	Address	Reset/default value	Test value
ENTRY_DELAY	0x008	DEF_ENTRY_DELAY = 30	30 + 10 = 40
EXIT_DELAY	0x00C	DEF_EXIT_DELAY = 30	30 + 20 = 50

3. Sequence Procedure

1. **Write ENTRY_DELAY:** Write DEF_ENTRY_DELAY + 10 (40) to address 0x008.
2. **Read ENTRY_DELAY:** Read address 0x008; expected readback is 40.
3. **Read EXIT_DELAY:** Read address 0x00C before changing it; expected value is the default 30.
4. **Write EXIT_DELAY:** Write DEF_EXIT_DELAY + 20 (50) to address 0x00C.
5. **Read ENTRY_DELAY again:** Expected value remains 40, confirming the EXIT_DELAY write did not alter ENTRY_DELAY.
6. **Read EXIT_DELAY again:** Expected value is 50.

4. Why the Sequence Uses +10 and +20

The offsets deliberately make the written values different from the default values. With defaults of 30, the sequence writes 40 to ENTRY_DELAY and 50 to EXIT_DELAY. If the test wrote the default value again, a readback of 30 would not clearly show whether the write had succeeded or the register had simply retained its reset value. The distinct values also help reveal if the two registers are swapped.

5. Expected Results

Operation	Expected value	Purpose
Write ENTRY_DELAY = 40; read it	40	Verify ENTRY_DELAY write/readback
Read EXIT_DELAY before its write	30	Verify default is retained
Write EXIT_DELAY = 50; read ENTRY_DELAY	40	Verify registers are independent
Read EXIT_DELAY after its write	50	Verify EXIT_DELAY write/readback

6. Observed Simulation Results

The supplied simulation log shows the following outcomes:

- The write to ENTRY_DELAY (0x008) used WDATA = 0x00000028 (decimal 40), but the subsequent read returned 0x00000000 (decimal 0). The AHB checker reported an ENTRY_DELAY mismatch.
- The initial read of EXIT_DELAY (0x00C) returned 0x0000001E (decimal 30), matching its default.
- The write to EXIT_DELAY used WDATA = 0x00000032 (decimal 50).
- The subsequent ENTRY_DELAY read returned 0x00000032 (decimal 50), rather than the expected 40.
- The final EXIT_DELAY read returned 0x0000001E (decimal 30), rather than the expected 50.

7. Result and Interpretation

The observed results are consistent with the known ENTRY_DELAY/EXIT_DELAY write-swap bug: writing 40 to ENTRY_DELAY is followed by ENTRY_DELAY reading as 0, and writing 50 to EXIT_DELAY is followed by ENTRY_DELAY reading as 50 while EXIT_DELAY remains at 30. The simulation log contains AHB checker errors for ENTRY_DELAY at the first and later reads, and for EXIT_DELAY at the final read. Therefore, this run does not pass.

8. Verification Note

The sequence shown performs writes and reads, but it does not itself compare read_seq.read_data against explicit expected values. The ahb_checker is reporting mismatches in the supplied log. For a sequence-level self-check, explicit comparisons should be added after each read, using expected values derived from DEF_ENTRY_DELAY and DEF_EXIT_DELAY.

9. Conclusion

The delay configuration sequence exercises both delay registers, tests their readback behavior, and is effective at exposing the register-write swap issue. The observed simulation run fails because the returned values do not match the intended values.

Appendix A — Simulation Log Evidence

Key log observations (times are from the supplied simulation output):

```
@24: AHB WRITE ADDR=00000008 WDATA=00000028 RESP=0

@60: ENTRY_DELAY mismatch: EXPECTED=00000028 ACTUAL=00000000

@96: EXIT_DELAY read RDATA=0000001e; EXIT_DELAY MATCH: VALUE=0000001e

@132: AHB WRITE ADDR=0000000c WDATA=00000032 RESP=0

@168: ENTRY_DELAY mismatch: EXPECTED=00000028 ACTUAL=00000032

@204: EXIT_DELAY mismatch: EXPECTED=00000032 ACTUAL=0000001e
```

Appendix B — Complete Simulation Log

The complete log supplied for this run is reproduced below. Error and mismatch lines are marked in red.

```
UVM_INFO @ 0: reporter [RNTST] Running test ahb_delay_config_test...
UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 18: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured:
ADDR=00000008 WRITE=1 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE
ADDR=00000008 WDATA=00000028 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing
transaction: ADDR=00000008 WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 24: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=00000008
HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(377) @ 24: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE:
ENTRY_DELAY expected=40
UVM_INFO scoreboard.sv(27) @ 24: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=00000008
DATA=00000028 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 24: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=00000008 | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_write_sequence.sv(28) @ 36: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_entry
[AHB_WRITE] WRITE ADDR=00000008 DATA=00000028 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 49: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 54: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured:
ADDR=00000008 WRITE=0 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 60: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB  READ
ADDR=00000008 WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 60: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing
transaction: ADDR=00000008 WRITE=0 RDATA=00000000
UVM_ERROR ahb_checker.sv(190) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ENTRY_DELAY mismatch:
EXPECTED=00000028 ACTUAL=00000000
UVM_INFO ahb_checker.sv(467) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB  READ: ADDR=00000008
HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 60: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=8
RDATA=0
UVM_INFO scoreboard.sv(27) @ 60: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB  READ: ADDR=00000008
DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 60: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=00000008 | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO sensor_monitor.sv(42) @ 63: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 63: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
```

```
UVM_INFO alarm_output_passive_monitor.sv(71) @ 65: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 72: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_entry [AHB_READ]
READ ADDR=00000008 RDATA=00000000 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 75: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 77: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 77: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 85: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 90: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured:
ADDR=0000000c WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 91: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 91: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 95: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 96: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB  READ
ADDR=0000000c WDATA=00000000 RDATA=0000001e HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 96: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing
transaction: ADDR=0000000c WRITE=0 RDATA=0000001e
UVM_INFO ahb_checker.sv(226) @ 96: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] EXIT_DELAY MATCH: VALUE=0000001e
UVM_INFO ahb_checker.sv(467) @ 96: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB  READ: ADDR=0000000c
HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 96: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=c
RDATA=1e
UVM_INFO scoreboard.sv(27) @ 96: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB  READ: ADDR=0000000c
DATA=00000000 RDATA=0000001e RESP=0
UVM_INFO sensor_checker.sv(157) @ 96: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=0000000c | WRITE=0 | RDATA=0000001e | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 105: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 105: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 105: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 108: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_exit [AHB_READ]
READ ADDR=0000000c RDATA=0000001e RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 115: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 119: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 119: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 125: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 126: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address
captured: ADDR=0000000c WRITE=1 HTRANS=10
UVM_INFO ahb_monitor.sv(104) @ 132: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE
ADDR=0000000c WDATA=00000032 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 132: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG]
Publishing transaction: ADDR=0000000c WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 132: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE:
ADDR=0000000c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(396) @ 132: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE:
EXIT_DELAY expected=50
UVM_INFO scoreboard.sv(27) @ 132: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=0000000c
DATA=00000032 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 132: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=0000000c | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO sensor_monitor.sv(42) @ 133: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 133: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 135: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_write_sequence.sv(28) @ 144: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_exit
[AHB_WRITE] WRITE ADDR=0000000c DATA=00000032 RESP=0
```

```
UVM_INFO alarm_output_passive_monitor.sv(71) @ 145: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 147: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 147: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 155: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 161: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 161: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 162: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address
captured: ADDR=00000008 WRITE=0 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 165: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 168: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB  READ
ADDR=00000008 WDATA=00000000 RDATA=00000032 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 168: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG]
Publishing transaction: ADDR=00000008 WRITE=0 RDATA=00000032
UVM_ERROR ahb_checker.sv(190) @ 168: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ENTRY_DELAY mismatch:
EXPECTED=00000028 ACTUAL=00000032
UVM_INFO ahb_checker.sv(467) @ 168: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB  READ:
ADDR=00000008 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 168: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=8
RDATA=32
UVM_INFO scoreboard.sv(27) @ 168: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB  READ: ADDR=00000008
DATA=00000000 RDATA=00000032 RESP=0
UVM_INFO sensor_checker.sv(157) @ 168: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=00000008 | WRITE=0 | RDATA=00000032 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 175: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 175: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 175: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 180: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_entry_again
[AHB_READ] READ ADDR=00000008 RDATA=00000032 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 185: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 189: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 189: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 195: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 198: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address
captured: ADDR=0000000c WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 203: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed
sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 203: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw
Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 204: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB  READ
ADDR=0000000c WDATA=00000000 RDATA=0000001e HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 204: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG]
Publishing transaction: ADDR=0000000c WRITE=0 RDATA=0000001e
UVM_ERROR ahb_checker.sv(217) @ 204: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] EXIT_DELAY mismatch:
EXPECTED=00000032 ACTUAL=0000001e
UVM_INFO ahb_checker.sv(467) @ 204: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB  READ:
ADDR=0000000c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 204: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=c
RDATA=1e
UVM_INFO scoreboard.sv(27) @ 204: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB  READ: ADDR=0000000c
DATA=00000000 RDATA=0000001e RESP=0
UVM_INFO sensor_checker.sv(157) @ 204: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION |
ADDR=0000000c | WRITE=0 | RDATA=0000001e | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 205: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 215: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG]
siren=0 strobe=0 alarm_active=0
```

```
UVM_INFO ahb_read_sequence.sv(31) @ 216: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_exit_again
[AHB_READ] READ ADDR=0000000c RDATA=0000001e RESP=0
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 216: reporter
[TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 216: reporter
[UVM/REPORT/SERVER]
--- UVM Report Summary --- &#x20;
```

AHB Register Access Test Report

Smart Home Alarm Controller — UVM Verification

1. Test Objective

Verify read access to selected software-visible registers through the AHB interface and compare the returned values against the expected reset/default values used by the testbench.

2. Registers Covered

Register	Address	Expected / Purpose	Observed Result
STATUS	0x004	Read current status; logged as 0	Read completed; no explicit sequence value comparison shown
ENTRY_DELAY	0x008	30 (0x1E)	FAIL — actual 0x00000000
EXIT_DELAY	0x00C	30 (0x1E)	PASS — actual 0x0000001E
ALARM_DURATION	0x010	180 (0xB4)	PASS — actual 0x000000B4
ZONE_ENABLE	0x018	0 after reset	PASS — actual 0x00000000
IRQ_STATUS	0x034	0 in this test	PASS — actual 0x00000000

3. Summary of Simulation

The test ran to completion at 216 ns. Six AHB read transactions were observed. All transactions reported HREADY=1, HREADYOUT=1, and HRESP=0 in the monitor output. The register checker and the sequence each reported an ENTRY_DELAY mismatch, so the UVM summary contains 2 errors and 0 fatal errors.

The other explicitly checked registers—EXIT_DELAY, ALARM_DURATION, ZONE_ENABLE, and IRQ_STATUS—matched their expected values. STATUS was read and logged, but this log does not show a dedicated expected-value comparison for STATUS.

4. Failure Analysis

Observed failure: ENTRY_DELAY at address 0x008 returned 0x00000000 instead of 0x0000001E (30).

The RTL source supplied for review contains reversed write assignments for ENTRY_DELAY and EXIT_DELAY. That is a known write-path defect, but this particular test performs reads only; therefore, this log alone does not establish that the swap caused the zero read. Record the result as a reset/default-read discrepancy in this test. No RTL modification was made for this report.

5. Test Conclusion

Status: TEST COMPLETED WITH FAILURE. Four of the five explicitly compared registers passed; ENTRY_DELAY failed. STATUS read completed but was not explicitly checked against an expected value in the sequence log. The failure is documented for tracking, and verification can proceed to the next planned test.

6. Complete Simulation Log

The full simulation log supplied for this run is reproduced below in order. No waveform is included, as requested.

```
UVM_INFO @ 0: reporter [RNTST] Running test ahb_register_access_test...
UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
```

UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 18: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000004 WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000004 WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000004 WRITE=0 RDATA=00000000
UVM_INFO ahb_checker.sv(172) @ 24: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] STATUS READ: ACTUAL=00000000; status validation delegated to dedicated checker
UVM_INFO ahb_checker.sv(467) @ 24: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000004 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 24: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=4 RDATA=0
UVM_INFO scoreboard.sv(27) @ 24: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000004 DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 24: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000004 | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 36: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_status [AHB_READ] READ ADDR=00000004 RDATA=00000000 RESP=0
UVM_INFO ahb_register_access_sequence.sv(20) @ 36: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [REG_ACCESS] STATUS = 0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 54: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000008 WRITE=0 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 60: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000008 WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 60: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000008 WRITE=0 RDATA=00000000
UVM_ERROR ahb_checker.sv(190) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ENTRY_DELAY mismatch: EXPECTED=0000001e ACTUAL=00000000
UVM_INFO ahb_checker.sv(467) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000008 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 60: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=8 RDATA=0
UVM_INFO scoreboard.sv(27) @ 60: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000008 DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 60: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000008 | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO sensor_monitor.sv(42) @ 63: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 65: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 72: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_entry_delay [AHB_READ] READ ADDR=00000008 RDATA=00000000 RESP=0
UVM_ERROR ahb_register_access_sequence.sv(29) @ 72: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [REG_ACCESS] ENTRY_DELAY mismatch: expected=1e actual=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 75: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 77: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 77: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 85: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 90: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=0000000c WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 91: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 95: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 96: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=0000000c WDATA=00000000 RDATA=0000001e HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 96: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=0000000c WRITE=0 RDATA=0000001e
UVM_INFO ahb_checker.sv(226) @ 96: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] EXIT_DELAY MATCH: VALUE=0000001e
UVM_INFO ahb_checker.sv(467) @ 96: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=0000000c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 96: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=c RDATA=1e
UVM_INFO scoreboard.sv(27) @ 96: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=0000000c DATA=00000000 RDATA=0000001e RESP=0
UVM_INFO sensor_checker.sv(157) @ 96: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=0000000c | WRITE=0 | RDATA=0000001e | RESP=0

UVM_INFO alarm_output_passive_monitor.sv(71) @ 105: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 105: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 108: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_exit_delay [AHB_READ] READ ADDR=0000000c RDATA=0000001e RESP=0
UVM_INFO ahb_register_access_sequence.sv(45) @ 108: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [REG_ACCESS] EXIT_DELAY PASSED
UVM_INFO alarm_output_passive_monitor.sv(71) @ 115: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 119: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 125: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 126: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000010 WRITE=0 HTRANS=10
UVM_INFO ahb_monitor.sv(104) @ 132: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000010 WDATA=00000000 RDATA=000000b4 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 132: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000010 WRITE=0 RDATA=000000b4
UVM_INFO ahb_checker.sv(253) @ 132: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ALARM_DURATION MATCH: VALUE=000000b4
UVM_INFO ahb_checker.sv(467) @ 132: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000010 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 132: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=10 RDATA=b4
UVM_INFO scoreboard.sv(27) @ 132: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000010 DATA=00000000 RDATA=000000b4 RESP=0
UVM_INFO sensor_checker.sv(157) @ 132: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000010 | WRITE=0 | RDATA=000000b4 | RESP=0
UVM_INFO sensor_monitor.sv(42) @ 133: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 135: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 144: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_alarm_duration [AHB_READ] READ ADDR=00000010 RDATA=000000b4 RESP=0
UVM_INFO ahb_register_access_sequence.sv(57) @ 144: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [REG_ACCESS] ALARM_DURATION PASSED
UVM_INFO alarm_output_passive_monitor.sv(71) @ 145: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 147: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 155: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 161: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_reference_model.sv(231) @ 161: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 162: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000018 WRITE=0 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 165: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 168: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000018 WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 168: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000018 WRITE=0 RDATA=00000000
UVM_INFO ahb_checker.sv(280) @ 168: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ZONE_ENABLE MATCH: VALUE=00000000
UVM_INFO ahb_checker.sv(467) @ 168: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000018 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 168: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=18 RDATA=0
UVM_INFO scoreboard.sv(27) @ 168: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000018 DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 168: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000018 | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 175: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 175: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 185: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 189: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 198: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000034 WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 203: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 204: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000034 WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 204: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000034 WRITE=0 RDATA=00000000
UVM_INFO ahb_checker.sv(361) @ 204: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] IRQ_STATUS MATCH: VALUE=00000000
UVM_INFO ahb_checker.sv(467) @ 204: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000034 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 204: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=34 RDATA=0
UVM_INFO scoreboard.sv(27) @ 204: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000034 DATA=00000000 RDATA=00000000 RESP=0

UVM_INFO sensor_checker.sv(157) @ 204: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000034 | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 205: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 216: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_irq_status [AHB_READ] READ ADDR=00000034 RDATA=00000000 RESP=0
UVM_INFO ahb_register_access_sequence.sv(81) @ 216: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [REG_ACCESS] IRQ_STATUS PASSED
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 216: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 216: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---
UVM_INFO : 113
UVM_WARNING : 0
UVM_ERROR : 2
UVM_FATAL : 0
[AHB_ADDRESS] 6
[AHB_CHECK] 12
[AHB_MON] 6
[AHB_PUBLISH_DEBUG] 6
[AHB_READ] 6

AHB Read/Write Sequence Verification Report

Smart Home Alarm Controller — UVM Verification

1. Objective

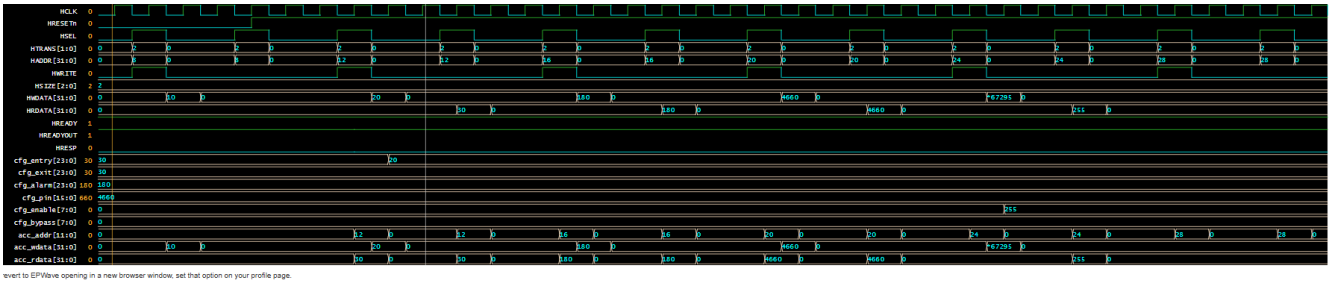
This report documents ahb_read_write_test, the current waveform, and simulation log evidence. The sequence writes to six memory-mapped registers, reads each back, and compares the returned data with expected values. Distinct values are used for ENTRY_DELAY and EXIT_DELAY to expose the injected register-swap bug. The RTL is not modified.

2. What the sequence does

- ENTRY_DELAY (0x008): writes TEST_ENTRY_DELAY = 0x0000000A, then expects readback 0x0000000A.
- EXIT_DELAY (0x00C): writes TEST_EXIT_DELAY = 0x00000014, then expects readback 0x00000014.
- ALARM_DURATION (0x010): writes DEF_ALARM_DUR = 0x000000B4 (180 decimal), then checks readback.
- PIN_CONFIG (0x014): writes DEF_PIN = 0x00001234, then checks readback.
- ZONE_ENABLE (0x018): writes all ones; because there are eight zones, expected readback is 0x000000FF.
- ZONE_BYPASS (0x01C): writes zero and expects zero on readback.
- Each register uses ahb_write_sequence followed by ahb_read_sequence; the readback is compared with the expected value.

3. Current waveform evidence

The waveform below replaces the previously included waveform. It shows HCLK, HRESETn, HSEL, HTRANS, HADDR, HWRITE, HSIZE, HWDATA, HRDATA, HREADY, HREADYOUT, HRESP, and internal configuration/access signals including cfg_entry, cfg_exit, cfg_alarm, cfg_pin, cfg_enable, cfg_bypass, acc_addr, acc_wdata, and acc_rdata.



4. Simulation log evidence

UVM_ERROR ahb_checker.sv(190) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ENTRY_DELAY mismatch: EXPECTED=0000000a ACTUAL=00000000

UVM_ERROR ahb_read_write_sequence.sv(39) @ 72: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ENTRY_DELAY FAILED: Expected=a Actual=0

UVM_ERROR ahb_checker.sv(217) @ 132: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] EXIT_DELAY mismatch: EXPECTED=00000014 ACTUAL=0000001e

UVM_ERROR ahb_read_write_sequence.sv(61) @ 144: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] EXIT_DELAY FAILED: Expected=14 Actual=1e

UVM_INFO ahb_read_write_sequence.sv(87) @ 216: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ALARM_DURATION PASSED

UVM_INFO ahb_read_write_sequence.sv(109) @ 288: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] PIN_CONFIG PASSED

UVM_INFO ahb_read_write_sequence.sv(132) @ 360: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ZONE_ENABLE PASSED

UVM_INFO ahb_read_write_sequence.sv(154) @ 432: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ZONE_BYPASS PASSED

--- UVM Report Summary ---

5. Results

Register	Address	Observed readback	Result
ENTRY_DELAY	0x008	Expected 0x0A; actual 0x00	FAIL
EXIT_DELAY	0x00C	Expected 0x14; actual 0x1E	FAIL
ALARM_DURATION	0x010	0xB4	PASS
PIN_CONFIG	0x014	0x1234	PASS
ZONE_ENABLE	0x018	0xFF	PASS
ZONE_BYPASS	0x01C	0x00	PASS

The log reports HRESP=0 and HREADYOUT=1 for the listed transfers, indicating successful responses. The ENTRY_DELAY and EXIT_DELAY readback mismatches are consistent with the injected RTL assignment swap: writes to ADDR_ENTRY_DELAY update exit_q, while writes to ADDR_EXIT_DELAY update entry_q. The read mux then returns the value stored in the opposite register.

6. Waveform signals to inspect

- AHB bus: HCLK, HRESETn, HSEL, HTRANS, HADDR, HWRITE, HSIZE, HWDATA, HRDATA, HREADY, HREADYOUT, HRESP.
- Register storage: entry_q, exit_q, alarm_q, pin_q, enable_q, bypass_q.
- Register access interface: acc_valid, acc_addr, acc_write, acc_wdata, acc_commit, acc_ready, acc_err, acc_rdata.
- Configuration outputs: cfg_entry, cfg_exit, cfg_alarm, cfg_pin, cfg_enable, cfg_bypass, cfg_wr.

7. Conclusion

The test exercises read/write behavior for six registers. ALARM_DURATION, PIN_CONFIG, ZONE_ENABLE, and ZONE_BYPASS pass readback checks. ENTRY_DELAY and EXIT_DELAY fail using distinct values, demonstrating detection of the injected swap bug. Keep the RTL unchanged and retain these failures as verification evidence.

Log color legend: RED with yellow highlight = UVM_ERROR / failed check; GREEN = passed or matched check; black = informational log.

8. Complete Simulation Log (Beginning to End)

The following is the complete simulation log from the uploaded text file, reproduced from its first line to its final line. Original message content and ordering are preserved.

UVM_INFO @ 0: reporter [RNTST] Running test ahb_read_write_test...
UVM_INFO alarm_output_passive_monitor.sv(71) @ 5: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 7: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 7: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 15: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 18: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000008 WRITE=1 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 21: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 21: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE ADDR=00000008 WDATA=0000000a RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 24: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000008 WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 24: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=00000008 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(377) @ 24: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE: ENTRY_DELAY expected=10
UVM_INFO scoreboard.sv(27) @ 24: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=00000008 DATA=0000000a RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 24: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000008 | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 25: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 35: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 35: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 35: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_write_sequence.sv(28) @ 36: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_entry_delay [AHB_WRITE] WRITE ADDR=00000008 DATA=0000000a RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 45: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 49: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 49: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 54: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000008 WRITE=0 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 55: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 60: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000008 WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 60: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000008 WRITE=0 RDATA=00000000
UVM_ERROR ahb_checker.sv(190) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ENTRY_DELAY mismatch: EXPECTED=0000000a ACTUAL=00000000
UVM_INFO ahb_checker.sv(467) @ 60: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000008 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 60: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=8 RDATA=0
UVM_INFO scoreboard.sv(27) @ 60: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000008 DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 60: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000008 | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO sensor_monitor.sv(42) @ 63: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 63: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 65: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 72: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_entry_delay [AHB_READ] READ ADDR=00000008 RDATA=00000000 RESP=0
UVM_ERROR ahb_read_write_sequence.sv(39) @ 72: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ENTRY_DELAY FAILED: Expected=a Actual=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 75: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 77: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 77: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 85: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 90: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=0000000c WRITE=1 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 91: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000

UVM_INFO alarm_reference_model.sv(231) @ 91: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 95: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 96: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE ADDR=0000000c WDATA=00000014 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 96: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=0000000c WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 96: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=0000000c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(396) @ 96: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE: EXIT_DELAY expected=20
UVM_INFO scoreboard.sv(27) @ 96: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=0000000c DATA=00000014 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 96: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=0000000c | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 105: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 105: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 105: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_write_sequence.sv(28) @ 108: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_exit_delay [AHB_WRITE] WRITE ADDR=0000000c DATA=00000014 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 115: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 119: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 119: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 125: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 126: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=0000000c WRITE=0 HTRANS=10
UVM_INFO ahb_monitor.sv(104) @ 132: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=0000000c WDATA=00000000 RDATA=0000001e HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 132: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=0000000c WRITE=0 RDATA=0000001e
UVM_ERROR ahb_checker.sv(217) @ 132: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] EXIT_DELAY mismatch: EXPECTED=00000014 ACTUAL=0000001e
UVM_INFO ahb_checker.sv(467) @ 132: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=0000000c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 132: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=c RDATA=1e
UVM_INFO scoreboard.sv(27) @ 132: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=0000000c DATA=00000000 RDATA=0000001e RESP=0
UVM_INFO sensor_checker.sv(157) @ 132: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=0000000c | WRITE=0 | RDATA=0000001e | RESP=0
UVM_INFO sensor_monitor.sv(42) @ 133: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 133: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 135: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 144: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_exit_delay [AHB_READ] READ ADDR=0000000c RDATA=0000001e RESP=0
UVM_ERROR ahb_read_write_sequence.sv(61) @ 144: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] EXIT_DELAY FAILED: Expected=14 Actual=1e
UVM_INFO alarm_output_passive_monitor.sv(71) @ 145: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 147: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 147: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 155: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 161: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 161: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 162: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000010 WRITE=1 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 165: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 168: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE ADDR=00000010 WDATA=000000b4 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 168: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000010 WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 168: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=00000010 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(415) @ 168: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE: ALARM_DURATION expected=180
UVM_INFO scoreboard.sv(27) @ 168: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=00000010 DATA=000000b4 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 168: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000010 | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 175: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 175: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 175: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000

UVM_INFO ahb_write_sequence.sv(28) @ 180: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_alarm_duration [AHB_WRITE] WRITE ADDR=00000010 DATA=000000b4 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 185: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 189: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 189: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 195: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 198: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000010 WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 203: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 203: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(104) @ 204: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000010 WDATA=00000000 RDATA=000000b4 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 204: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000010 WRITE=0 RDATA=000000b4
UVM_INFO ahb_checker.sv(253) @ 204: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ALARM_DURATION MATCH: VALUE=000000b4
UVM_INFO ahb_checker.sv(467) @ 204: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000010 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 204: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=10 RDATA=b4
UVM_INFO scoreboard.sv(27) @ 204: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000010 DATA=00000000 RDATA=000000b4 RESP=0
UVM_INFO sensor_checker.sv(157) @ 204: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000010 | WRITE=0 | RDATA=000000b4 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 205: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 215: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_read_sequence.sv(31) @ 216: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_alarm_duration [AHB_READ] READ ADDR=00000010 RDATA=000000b4 RESP=0
UVM_INFO ahb_read_write_sequence.sv(87) @ 216: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ALARM_DURATION PASSED
UVM_INFO sensor_monitor.sv(42) @ 217: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 217: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 225: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 231: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 231: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 234: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000014 WRITE=1 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 235: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 240: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE ADDR=00000014 WDATA=00001234 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 240: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000014 WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 240: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=00000014 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(453) @ 240: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE: PIN_CONFIG expected=1234
UVM_INFO scoreboard.sv(27) @ 240: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=00000014 DATA=00001234 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 240: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000014 | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 245: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 245: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 245: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_write_sequence.sv(28) @ 252: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_pin_config [AHB_WRITE] WRITE ADDR=00000014 DATA=00001234 RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 255: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 259: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 259: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 265: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 270: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000014 WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 273: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 273: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 275: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 276: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000014 WDATA=00000000 RDATA=00001234 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 276: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000014 WRITE=0 RDATA=00001234
UVM_INFO ahb_checker.sv(467) @ 276: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000014 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1

UVM_INFO alarm_reference_model.sv(565) @ 276: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=14 RDATA=1234
UVM_INFO scoreboard.sv(27) @ 276: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000014 DATA=00000000 RDATA=00001234 RESP=0
UVM_INFO sensor_checker.sv(157) @ 276: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000014 | WRITE=0 | RDATA=00001234 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 285: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 287: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 287: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 288: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_pin_config [AHB_READ] READ ADDR=00000014 RDATA=00001234 RESP=0
UVM_INFO ahb_read_write_sequence.sv(109) @ 288: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] PIN_CONFIG PASSED
UVM_INFO alarm_output_passive_monitor.sv(71) @ 295: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 301: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 301: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 305: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 306: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000018 WRITE=1 HTRANS=10
UVM_INFO ahb_monitor.sv(104) @ 312: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE ADDR=00000018 WDATA=ffffffff RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 312: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000018 WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 312: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=00000018 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(472) @ 312: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE: ZONE_ENABLE expected=ff
UVM_INFO scoreboard.sv(27) @ 312: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=00000018 DATA=ffffffff RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 312: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000018 | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 315: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 315: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 315: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_write_sequence.sv(28) @ 324: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_zone_enable [AHB_WRITE] WRITE ADDR=00000018 DATA=ffffffff RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 325: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 329: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 329: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 335: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 342: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=00000018 WRITE=0 HTRANS=10
UVM_INFO sensor_monitor.sv(42) @ 343: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 343: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 345: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 348: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=00000018 WDATA=00000000 RDATA=000000ff HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 348: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=00000018 WRITE=0 RDATA=000000ff
UVM_INFO ahb_checker.sv(280) @ 348: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ZONE_ENABLE MATCH: VALUE=000000ff
UVM_INFO ahb_checker.sv(467) @ 348: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=00000018 HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 348: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=18 RDATA=ff
UVM_INFO scoreboard.sv(27) @ 348: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=00000018 DATA=00000000 RDATA=000000ff RESP=0
UVM_INFO sensor_checker.sv(157) @ 348: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=00000018 | WRITE=0 | RDATA=000000ff | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 355: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 357: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 357: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 360: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_zone_enable [AHB_READ] READ ADDR=00000018 RDATA=000000ff RESP=0
UVM_INFO ahb_read_write_sequence.sv(132) @ 360: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ZONE_ENABLE PASSED
UVM_INFO alarm_output_passive_monitor.sv(71) @ 365: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 371: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 371: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 375: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(65) @ 378: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=0000001c WRITE=1 HTRANS=10
UVM_INFO ahb_monitor.sv(104) @ 384: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB WRITE ADDR=0000001c WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0

UVM_INFO ahb_monitor.sv(121) @ 384: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=0000001c WRITE=1 RDATA=00000000
UVM_INFO ahb_checker.sv(467) @ 384: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB WRITE: ADDR=0000001c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(491) @ 384: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB WRITE: ZONE_BYPASS expected=0
UVM_INFO scoreboard.sv(27) @ 384: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB WRITE: ADDR=0000001c DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 384: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=0000001c | WRITE=1 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 385: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 385: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 385: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 395: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_write_sequence.sv(28) @ 396: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@write_zone_bypass [AHB_WRITE] WRITE ADDR=0000001c DATA=00000000 RESP=0
UVM_INFO sensor_monitor.sv(42) @ 399: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 399: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO alarm_output_passive_monitor.sv(71) @ 405: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 413: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 413: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_monitor.sv(65) @ 414: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_ADDRESS] Address captured: ADDR=0000001c WRITE=0 HTRANS=10
UVM_INFO alarm_output_passive_monitor.sv(71) @ 415: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO ahb_monitor.sv(104) @ 420: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_MON] AHB READ ADDR=0000001c WDATA=00000000 RDATA=00000000 HREADY=1 HREADYOUT=1 RESP=0
UVM_INFO ahb_monitor.sv(121) @ 420: uvm_test_top.alarm_env.ahb_agent.ahb_monitor [AHB_PUBLISH_DEBUG] Publishing transaction: ADDR=0000001c WRITE=0 RDATA=00000000
UVM_INFO ahb_checker.sv(307) @ 420: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] ZONE_BYPASS MATCH: VALUE=00000000
UVM_INFO ahb_checker.sv(467) @ 420: uvm_test_top.alarm_env.ahb_chk [AHB_CHECK] Checked AHB READ: ADDR=0000001c HTRANS=10 HSIZE=10 RESP=0 HREADYOUT=1
UVM_INFO alarm_reference_model.sv(565) @ 420: uvm_test_top.alarm_env.ref_model [REF_MODEL] AHB READ: ADDR=1c RDATA=0
UVM_INFO scoreboard.sv(27) @ 420: uvm_test_top.alarm_env.scoreboard [SCOREBOARD] AHB READ: ADDR=0000001c DATA=00000000 RDATA=00000000 RESP=0
UVM_INFO sensor_checker.sv(157) @ 420: uvm_test_top.alarm_env.sensor_chk [SENSOR_AHB_DEBUG] AHB TRANSACTION | ADDR=0000001c | WRITE=0 | RDATA=00000000 | RESP=0
UVM_INFO alarm_output_passive_monitor.sv(71) @ 425: uvm_test_top.alarm_env.alarm_output_mon [ALARM_DEBUG] siren=0 strobe=0 alarm_active=0
UVM_INFO sensor_monitor.sv(42) @ 427: uvm_test_top.alarm_env.sensor_agent.sensor_monitor [SENSOR_MON] Observed sensor_in = 00000000
UVM_INFO alarm_reference_model.sv(231) @ 427: uvm_test_top.alarm_env.ref_model [REF_MODEL] SENSOR: Raw Input=00000000 Sync Sample=00000000 Expected Zone Level=00000000
UVM_INFO ahb_read_sequence.sv(31) @ 432: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@read_zone_bypass [AHB_READ] READ ADDR=0000001c RDATA=00000000 RESP=0
UVM_INFO ahb_read_write_sequence.sv(154) @ 432: uvm_test_top.alarm_env.ahb_agent.ahb_sequencer@@seq [AHB_READ_WRITE] ZONE_BYPASS PASSED
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 432: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO /apps/vcsmx/vcs/X-2025.06-SP1//etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 432: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---